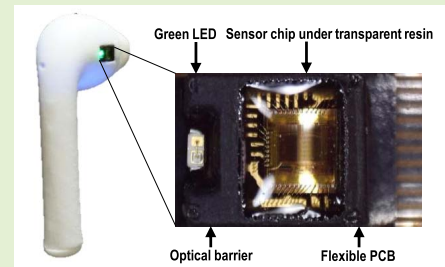


Earbud-Embedded Micro-Power mm-Sized Optical Sensor for Accurate Heart Beat Monitoring

Assim Boukhayma¹, Anthony Barison, Serj Haddad², and Antonino Caizzone

Abstract—This work demonstrates that accurate heart beat monitoring, via photoplethysmography (PPG), can be performed at micro-power consumption level with an ear worn device. This is achieved thanks to the combination of a highly sensitive low power PPG chip, an optimal optical module, and an ergonomic design of the wearable device. The monolithic optical sensor is fabricated in a 180 nm CMOS image sensor (CIS) process and features an area of 1.5 mm by 1.5 mm only. It implements an array of pixels, newly designed for PPG, and reaching a quantum efficiency of 85% up to 800 nm. The presented sensor consumes 60 μ A at 122Hz sampling frequency, including digital communication. The optical design optimization enables high signal-to-noise ratio (SNR) PPG signals with less than 10 μ A average LED current. Heart beat monitoring, using the presented wearable device, is performed on 7 subjects and compared with a medical electrocardiogram (ECG). The earbud device correctly detects 98.47% of the beats on a total of 72.21 hours of recordings. The inter-beat interval (IBI) estimation from these recordings features a mean absolute error (MAE) as low as 7.52ms.

Index Terms—PPG, PPD, ultra-low-power, CMOS, sensor, IBI, wearable.



I. INTRODUCTION

NOWADAYS, efficient and remote health monitoring is becoming increasingly important, given both the ageing of population and the combined action of an increase in obesity level and cardiovascular diseases. The healthcare industry is becoming more reliant on new methods to monitor and treat patients. This, along with an increased interest in fitness and wellness, is calling for more affordable, precise and wearable health monitoring devices. In this context, photoplethysmography (PPG), also named optical heart sensing, appears to be a key technology allowing non-invasive monitoring of vital signs such as the heart rate, the blood oxygen saturation, the respiration rate and the arterial blood pressure [1].

A PPG signal is obtained by shining light from one or more LEDs at a given wavelength, from visible to near-

infrared, into a human tissue, e.g. finger, wrist, ear lobes. As shown in Fig. 1, a photodetector (PD) detects the light transmitted through or reflected from the tissue and transforms it into a photogenerated current. The detected signal, i.e. PPG, consists of two different components: a large DC (quasi-static) component corresponding to the light diffusion through tissues and non-pulsatile blood layers, and a small AC (pulsatile) part due to the diffusion through the arterial blood. The AC component is only a very small fraction (typically 0.2% to 10%) of the DC one, depending on the body location and the skin tone [2]. For instance, at the same power, a wrist PPG exhibits a fairly lower AC than a finger one. The AC/DC ratio is called Perfusion-Index (PI) and represents one of the challenges of any PPG system in terms of dynamic range (DR) and power consumption [1], given that the majority of the medical information is carried by the tiny AC component.

The use of PPG sensors in wearables has been at first dominated by the smartwatches segment. Lately, the earbud segment is gaining the interest of most consumer wellness and health electronic devices manufacturers. The earbuds are very suitable for several use cases from fitness tracking to sleep monitoring. Moreover, the body location of the PPG sensor is an important issue affecting the signal quality or the sensor resilience to motion artifacts. In this regard, the ear is considered as one of the best locations in terms of PPG signals. It has been shown that the ear (particularly the canal area,

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This work involved human subjects or animals in its research. All ethical and experimental procedures and protocols were performed in line with the guidelines of the Helsinki Declaration of 1975, as revised in 2000, and the subjects provided an informed consent to participate in the study and could have withdrawn for the study at any time. Prior the analysis, all data are rendered anonymous.

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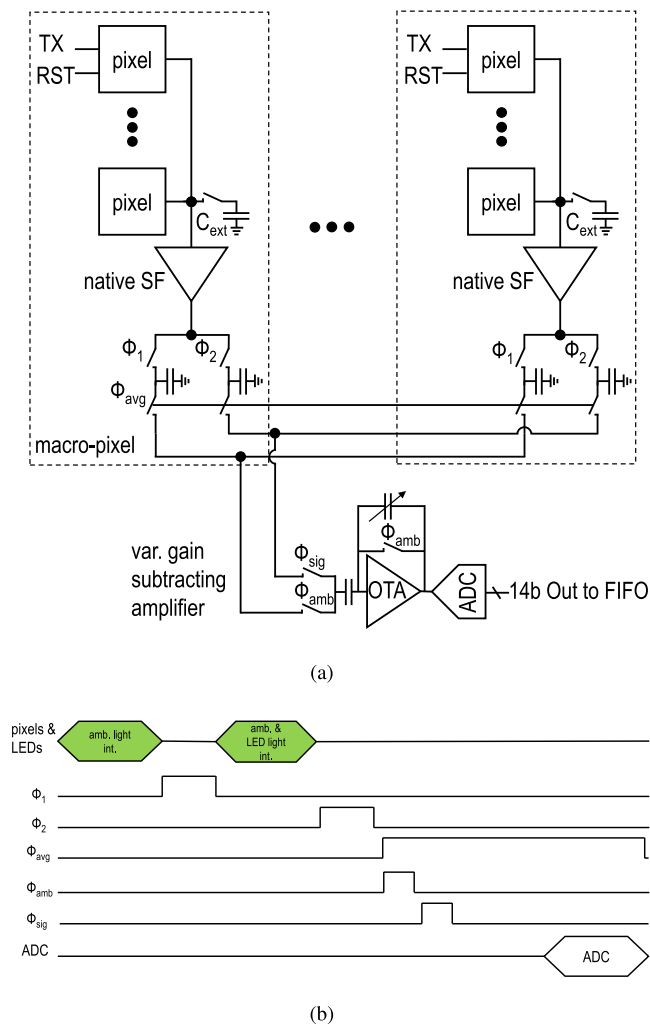


Fig. 3. (a) A block diagram of the sensor depicting the macro-pixels array, the subtracting amplifier and ADC, and (b) corresponding timing diagram.

connected to the column to perform charge binning during the readout. A reset is performed before each integration. This allows to empty the PPD well and, at the same time, to set the column at a high voltage in order to attract the electrons accumulated in the PPD well during the integration. The pixels can perform two consecutive light integrations during windows as short as $10\mu s$. This allows two ambient light resilience mechanisms. The first is based on the short integration time and the second on the double sampling scheme that allows cancellation of the residual light. The pixel also offers an important advantage for power reduction, on the emitter side, from the QE perspective. Indeed, a higher sensitivity leads to a same signal-to-noise ratio (SNR) with less emitted light. The presented pixel QE is improved thanks to the introduction of a weakly doped epitaxial layer under the PPD well. This weakly doped layer enhances the depth of the PPD depleted area which results in increasing the survival probability of electron-hole pairs, especially the ones generated deeper in the silicon, corresponding mostly to the higher wavelengths photons.

As shown in Fig. 3, the pixels sharing the same column SN are embedded in macro-pixels (MPs). Each MP also

encompasses a switchable excess capacitance that enhances the dynamic range and a native source follower (SF) stage, operated in weak inversion region to ensure low power operations. The SF converts the charge transferred from the PPD to the SN into a voltage. Two sample-and-hold capacitors at the output of the SF ensure a double sampling scheme: one sample for the ambient light transfer and the second for the LED-originating light superimposed on the ambient one as described earlier in Fig. 2. Both samples are then averaged over the whole array in order to mitigate the electronic noise generated by the active circuits of the MPs. Indeed, it has been shown in [1] that an array of PPDs guarantees an important noise and power optimization. The two averaged samples are then fed to a switched capacitor (SC) variable gain amplifier stage that subtracts the two samples and buffers the amplified difference to an incremental 14 bit ADC. The working principle of the ADC and SC amplifier are detailed in [12]. Fig. 3(b) reports the whole front-end timing diagram, from the light integration to the ADC.

The presented monolithic sensor chip is designed to cope with a dynamic range of 90dB, in order to cope with the high DC component of the PPG signal. The pixels feature a well density of $3 \cdot 10^5$ electrons/pixel. The number of photo-electrons necessary to reach 90dB SNR is approximately $1.01376 \cdot 10^9$, hence the presented sensor features an array of 4096 pixels equally split between the 128 MPs. Regarding the readout chain, the native SFs stages ensure high dynamic since they feature a threshold voltage close to zero and hence an output swing of more than 2V. At the level of the SN, the switchable capacitance C_{ext} allows to extend the dynamic in case the charge transferred to the SN induces a voltage drop exceeding 2V. The amplifier is designed to feature a 2.5V swing to preserve the dynamic range before the ADC.

The global architecture of the sensor analog readout chain is depicted in Fig. 4, together with the corresponding timing diagram. The chip also embeds, in addition to the pixels array and associated readout chain, LED drivers, embedded oscillators, digital control, 128 FIFO registers, and a serial peripheral interface (SPI) communication unit.

In order to optimize the power consumption at the circuit level, the sequential nature design of the readout chain is exploited extensively. As shown in Fig. 4(b), the array of SFs operating in weak inversion are disconnected from the power line once the ambient light and signal levels are stored in the MP capacitors. After that, the amplifier and ADC are powered only during the conversion. All of the chip blocks can operate with a slow clock of 500 kHz, except the incremental ADC that requires at least 50 MHz to obtain enough resolution in a reasonable time. Hence, the chip embeds two oscillators: a slow one consuming less than $2\mu A$ and always on, and a second, oscillating at 50 MHz which is only turned on during the ADC operation. The digital design is also split in two blocks: one operating with the slow clock to ensure all of the chip functions and a sub-block operating with the fast clock only during the ADC operation. In this way, the chip power consumption is optimized taking advantage of both power and clock gating.

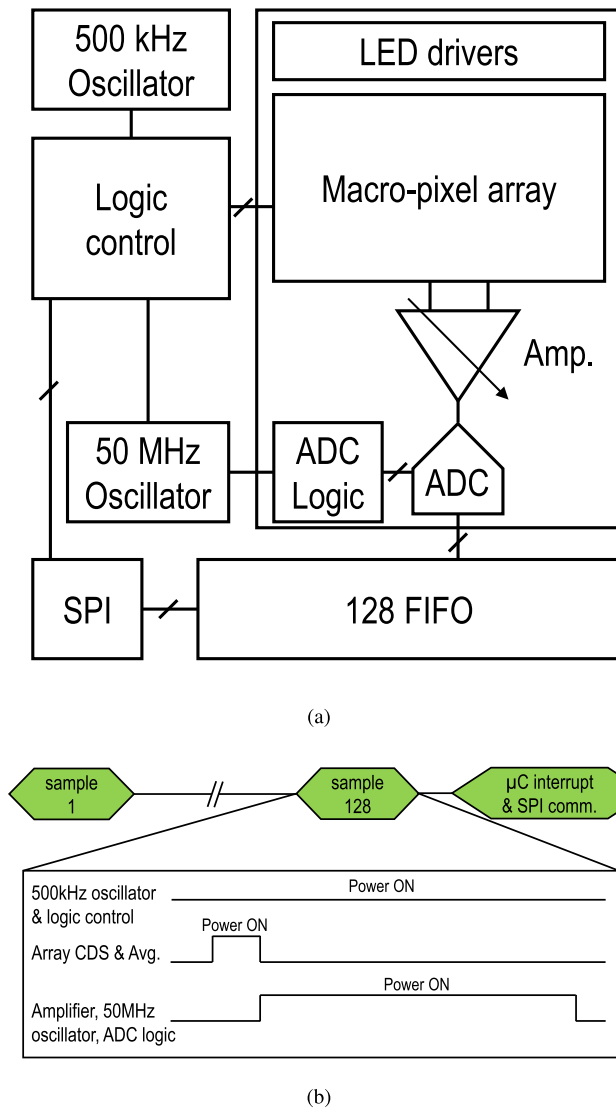


Fig. 4. (a) A block diagram of the sensor chip system design depicting the main functional blocks, and (b) corresponding timing diagram depicting the power and clock gating mechanisms.

B. Sensor Photo-Electrical Test

Fig. 5 shows a micrograph picture of the dedicated monolithic chip, specifically designed for ear PPG applications. The chip is fabricated in a 180 nm CIS process. The main blocks of the chip are highlighted in Fig. 5, namely, the dedicated pixels array, the amplifier, the ADC, the LED drivers, the embedded oscillators, digital control, 128 FIFO registers, and an SPI communication unit. Fig. 6 shows the measured QE of two pixel variants featuring two weak P type epitaxial layer depths. The proposed pixel features a weak P epitaxial depth of $12\mu\text{m}$, while the comparison pixel presented in [11], [12] features a $5\mu\text{m}$ epitaxial layer. The weakly doped epitaxial layer role helps to deepen the depleted area of the PPD device. As already mentioned in Section II-A, this results in increasing the probability of electron-hole pairs generated deeper in the PPD to survive. This mechanism increases the QE especially towards near-infra-red (NIR) spectra, as shown by Fig. 6.

Fig. 7 shows the SNR, measured at the sensor output raw data (without digital filtering), in loop-back mode over the full

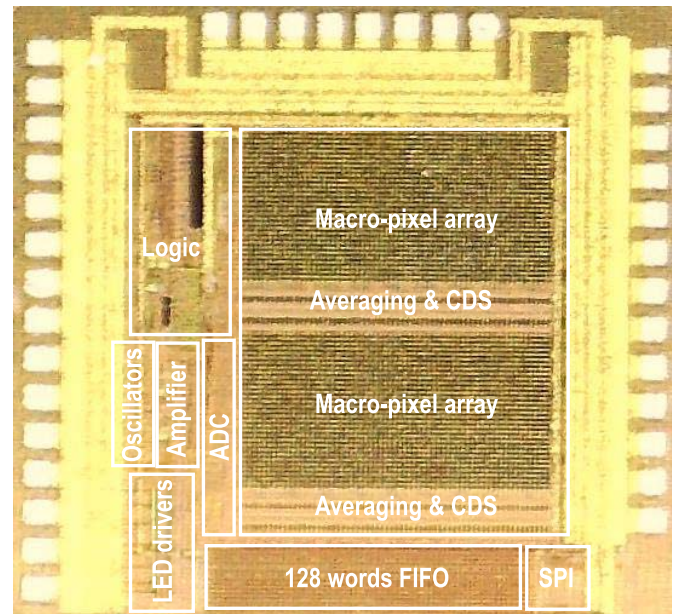


Fig. 5. Die micrograph showing the 1.5mm by 1.5mm sensor main building blocks integrated with pixels array.

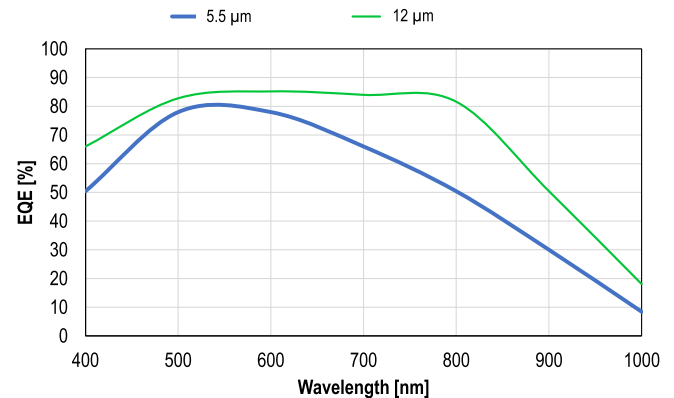


Fig. 6. Measured EQE of two versions of the PPD based pixel dedicated for PPG with two weakly doped epitaxial layer depth.

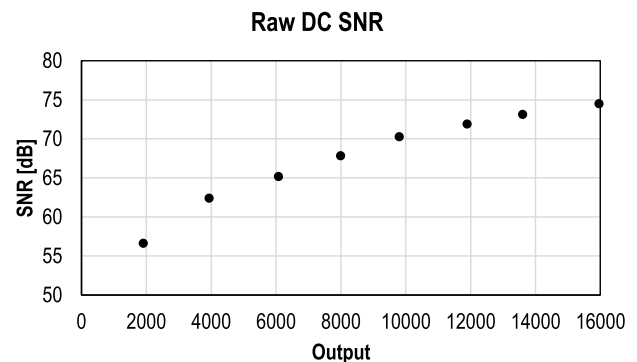


Fig. 7. Measured SNR as a function of the sensor output obtained by exposing the sensor to a varying input reflected DC light in loop-back mode.

DR of the sensor. This measurement is obtained by exposing the module, composed of the sensor and LEDs, to a stable reflector. The points of the curve are obtained by gradually increasing the LED driving current (set by the chip) and

TABLE I
SENSOR CHARACTERISTICS

Parameter	This work	[14]	[15]	[16]	[17]
Monolithic	yes	no	yes	no	no
Process	180 nm CIS	180 nm	65 nm CIS	180 nm	55 nm
Voltage supply	3.3/1.8 V	3.3/1.2 V	1.8/1 V	3.3/1.2 V	2.8/1.8 V
Area	2.25 mm ²	7 mm ² excluding PD	5.5 mm ²	8.37 mm ² excluding PD	4.5 mm ² excluding PD
Integrated FIFO and SPI/I2C	yes (128 words)	no	yes (64 words)	no	yes with 16 kB memory
Max. sampling frequency	500 Hz	2048 Hz	20 Hz	2048 Hz	20 Hz
Max. dynamic range	90 dB	119 dB	90 dB	134 dB	130 dB
ADC resolution	14 bit	14 bit	16 bit	10 bit	14 bit
DR enhancement technique	extra SN capacitance	7 bit DC cancellation	High ADC resolution	7 bit DC cancellation and noise shaping	11 bit DC cancellation
Dynamic ambient light rejection	84 dB @ 1 Hz with 20 dB per decade decrease	not reported	46 dB below 120 Hz	not reported	not reported
Consumption	60 μ A @ 122 Hz including digital	74 μ A @ 522 Hz excluding digital	24 μ A @ 20 Hz including digital	28 μ W	72 μ W

measuring the SNR for each LED driving configuration. In this way, this measurement of the SNR includes, not only the noise generated in the detection and readout process, but also the noise of the LED drivers and the LED themselves. The full system can go up to 75 dB without digital filtering. Since a PPG signal is very often digitally filtered during the processing, the maximum SNR increases to 85 dB after band limiting the signal between 0.1 Hz and 4 Hz, thanks to a second order Butterworth filter. The maximum SNR is obtained when pixel array accumulates a minimum of 64 million photo-electrons, corresponding to an input photo-current of 115 nA for an integration window of 100 μ s.

Ambient light resilience is an important characteristic of PPG sensors. The ambient light attenuation of the presented sensor is measured for both static and dynamic light. The measurement setup consists of placing the sensor in a dark chamber directly exposed to a wide band light emitter. The emitter is set to an intensity that reaches the saturation level of the sensor when the ambient light cancellation mode of the sensor is disabled. The attenuation is measured then as the sensor output when the ambient light cancellation is activated, normalized to the sensor maximum output. When operating the emitter in DC mode, the sensor cancels completely the ambient light. When the emitter is modulated at 1 Hz, an attenuation of 84 dB is reached. Fig. 8 shows the measured attenuation, when the emitter is modulated up to 1 kHz. The attenuation decreases following a 20 dB per decade slope, starting with 65 dB at 10 Hz.

Tab. I shows a summary of the main chip characteristics and performance metrics in comparison with recent SOA. The introduced chip presents a monolithic solution that stands out in terms of silicon area, power consumption and ambient light rejection, while embedding a 128 FIFO register and SPI communication protocol. The presented chip exhibits a relatively lower DR with respect to [14], [16], [17]. Over 120 dB DR is achieved in the reported literature thanks to a feedback loop subtracting a DC current component from the PD node at the input of the readout chain. Such a DC

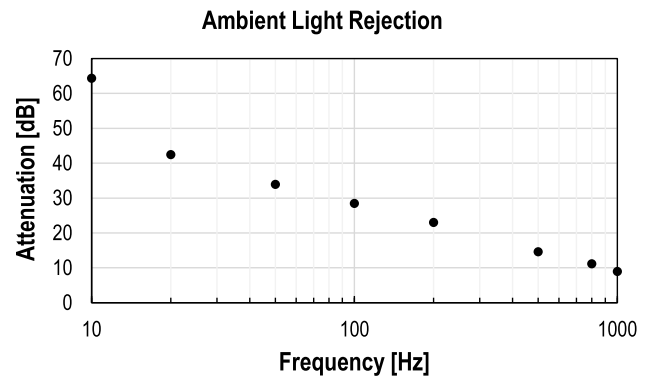
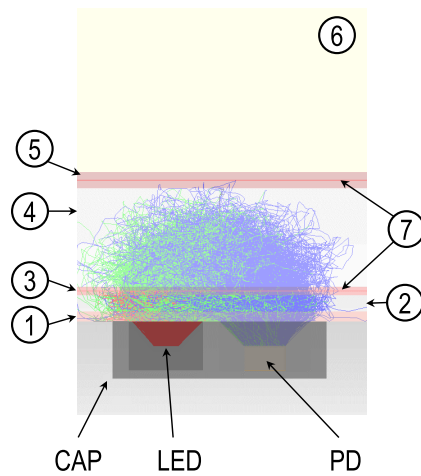


Fig. 8. Dynamic ambient light attenuation measurement.

cancellation mechanism can certainly increase the DR, but also shows weakness with respect to dynamic ambient light. Indeed, against a fast ambient light variation, the DC cancellation loop might not react fast enough, leading to saturation. Large DR is particularly needed for PPG in body locations featuring low PI such as chest or forehead. The presented chip is rather targeting the ear concha or tragus, that feature a PI usually higher than 1%. Hence 90 dB dynamic is considered large enough for such application, especially that, for ear PPG, the LED illumination must stay as low as possible in order to remain invisible (for green light) and cope with the power consumption strict constraints. However, the proposed readout scheme can also be complemented by two DR enhancement techniques. The first consists of switchable extra capacitances at the SN. The second consists of performing a series of short exposures instead of a single one as shown in Fig. 2 and averaging the outputs in the same way the spatial averaging is performed. Successive short exposure can prevent the PPD and SN from saturation in case of large input light.

III. OPTICAL MODULE OPTIMIZATION

The optimization of a PPG module addresses several physical domains, such as optics, mechanics, electronics, and biology. In this section, the optimization of the opto-mechanical



No.	Material	Thick. [μm]	Refr. Index	Absorption [1/mm]	*Anisotropy (g)	*Scatter Coeff. [1/mm]
1	Epidermis	120	1.335	0.5 @ 530 nm 0.26 @ 660 nm	0.8 @ 530 nm 0.8 @ 660 nm	31.3 @ 530 nm 22.1 @ 660 nm
2	Papillary dermis	200	1.37	0.28 @ 530 nm 0.15 @ 660 nm	0.8 @ 530 nm 0.8 @ 660 nm	19.2 @ 530 nm 14.4 @ 660 nm
3	Plexus superficialis	100	1.4	1.20 @ 530 nm 0.33 @ 660 nm	0.78 @ 530 nm 0.82 @ 660 nm	21.9 @ 530 nm 19.2 @ 660 nm
4	Reticular dermis	1200	1.37	0.28 @ 530 nm 0.15 @ 660 nm	0.8 @ 530 nm 0.8 @ 660 nm	19.2 @ 530 nm 14.4 @ 660 nm
5	Plexus profundus	200	1.4	1.38 @ 530 nm 0.34 @ 660 nm	0.78 @ 530 nm 0.82 @ 660 nm	22.5 @ 530 nm 19.4 @ 660 nm
6	Hypodermis	5000	1.44	0.28 @ 530 nm 0.4 @ 660 nm	0.8 @ 530 nm 0.8 @ 660 nm	16.3 @ 530 nm 12.3 @ 660 nm
7	Blood	Var.	1.335	180 @ 530 nm 1.88 @ 660 nm	0.8 @ 530 nm 0.8 @ 660 nm	701 @ 530 nm 894 @ 660 nm

* Henyey-Greenstein phase function.

Fig. 9. Skin model and optical parameters.

coupling between the PPG system and the skin is studied by simulating the effect of the distance between the LED and the PD array in terms of PI, SNR and optical power.

As discussed in [1], a PPG signal consists of a large DC component and a small AC one imposed upon the former. The AC/DC ratio, namely PI, the larger the better, is the most important parameter around a PPG signal. First of all, a skin model was developed based on the literature. Histological studies, such as [18]–[22], finely describe the anatomical structure of the skin, in which three macro layers can be identified, namely the epidermis, the dermis and the hypodermis. The optical properties of the anatomical structure of the skin can be modeled by a seven layer structure, that was proven sufficient to model primary effects of photon-skin interaction, such as absorption and scattering [23]–[25]. Fig. 9 shows a cross-sectional cut-view of the simulated system consisting of a skin model, a simplified PPG system and the photon-path scattered through the skin. The optical properties and thickness of each layer were adapted from the literature, [20]–[22], [25]. It is worth noting that the statistical nature of such values is subject to laboratory procedures, sample preparation, origin of the sample. On the other hand, the ratio between epidermis and overall dermis as well as the percentage volume of blood for a given volume of dermis carries greater significance in understanding signal quality for a given optical flux. It is key, for whatever optical simulation intended to maximize the AC/DC ratio, to present a correct blood volume model, which ultimately accounts for its dynamic properties. In this regard, pulsatile component of blood volume was modeled as two homogeneous layers of oxygenated blood in the DPS and DPP regions having the following characteristics: a volumetric ratio of 1 : 2 and an overall volume of 0.6% (systole) and 0.2% (diastole) of the overall physical volume of dermis [26].

TracePro® and macro-scripting were used to combine Monte Carlo simulations and repetitive raytracing at differing blood volumes, while sweeping the distance between the centres of the PPG chip (PD) and LED. Fig. 10 shows the simulated relation between relative PI and PD-LED distance (Δy). The PI increases with the PD-LED distance. In agreement

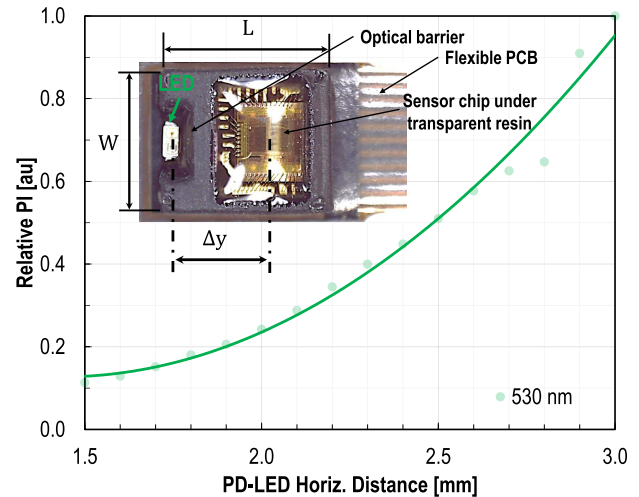


Fig. 10. Simulated relation between relative PI and PD-LED distance at 530 nm.

with the Beer-Lambert's Law, the DC component is found to decrease exponentially at a faster rate relative to the AC component.

The PPG signal SNR at the input of the optical sensor (PD) depends exclusively on the PI and the received optical power. Indeed, under the assumption that the shot noise is the dominant noise contributor and a low PI, the SNR can be approximated by $20 \log_{10}(PI \cdot \sqrt{N})$, where N is the number of detected photons. Fig. 11 shows the simulated SNR as function of the LED-sensor distance for a given emitted optical power. As expected by the above SNR estimation, the simulation shows that even though the PI increases with the LED-sensor distance, the SNR degrades due to the exponential decay of the number of received photons. Hence, from a pure SNR perspective, the LED-sensor distance must be kept as short as possible.

From a practical design aspect, a short LED-sensor distance also means small PI and hence very tight constraints on the dynamic range of the readout chain. A high dynamic readout chain requires large photo-electron capacity, high ADC

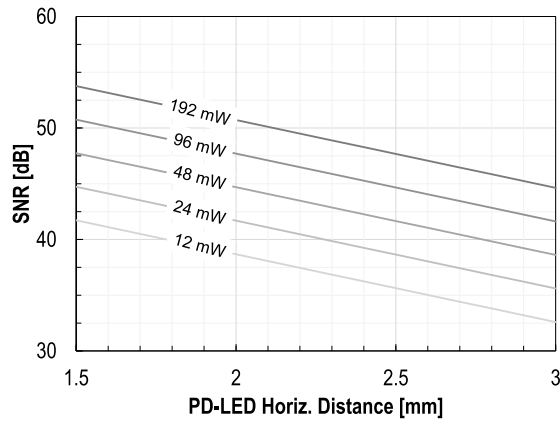


Fig. 11. Simulated relation between the SNR at the input of the sensor and PD-LED distance with 530 nm LED illumination.

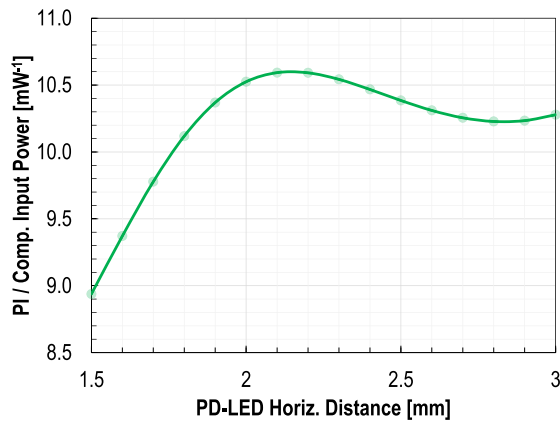


Fig. 12. Simulated relation between the PI normalized to the LED power and PD-LED distance with 530 nm illumination and a constant SNR.

resolution and tight noise constraints, which inevitably results in high power consumption and silicon area. In order to include the design complexity in our analysis, we define a figure-of-merit as the ratio between the PI and the optical power. Fig. 12 shows the simulated PI normalized to the optical power consumption for a constant SNR of 42 dB. This simulation shows that an optimum between the design complexity and optical power consumption exists at about 2.2 mm LED-sensor distance.

The optical simulations enable the optimization of the PPG module, consisting of the presented sensor and LED, shown in the inset of Fig. 10. The LED is an OSRAM CT DELSS1.12 with nominal wavelength at 530 nm. The module size ($W \cdot L \cdot H$) is $3.2 \cdot 3.8 \cdot 0.6 \text{ mm}^3$. The L dimension is ultimately set by the choice of the centre-to-centre distance between the PPD pixels array and the LED. We decided to place the LED at $\Delta y = 2.2 \text{ mm}$ based on the above mentioned analysis.

IV. EARBUD INTEGRATION

The PPG module was embedded into an adhoc wearable earbud. The integration of the system faces a few challenges, namely, the miniaturization of the electronic system and the ergonomics. Fig. 13 shows top and planar AA view of

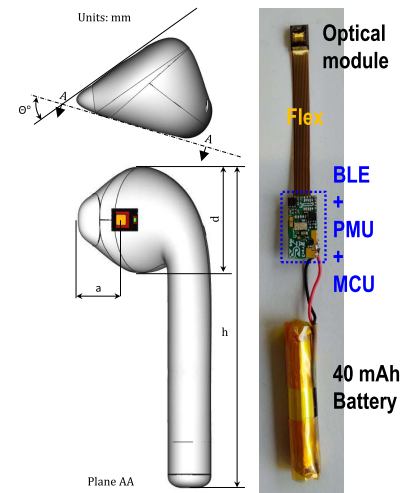


Fig. 13. Top and planar AA view of the earbud, including tear-down of the electronic system.

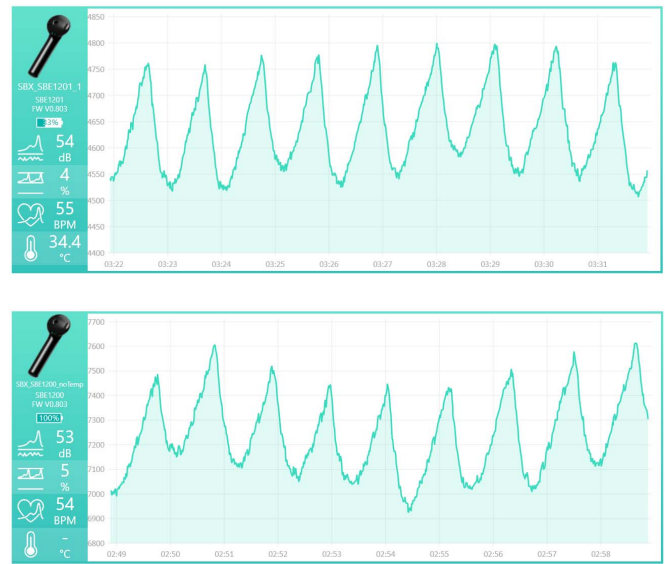
the design, and a photo of the tear-down of the system showing the PPG module connected to the microcontroller unit and a set of custom-shaped batteries. The electronic system consists of the PPG module, a PCB hosting the microcontroller (MCU), power management unit (PMU) and low energy Bluetooth (BLE) and a 40 mAh cylindrical battery, 15 mm by 5 mm. The small form factor of the PPG module in Fig. 10 was beneficial in the positioning of the sensor to follow the curvature of the outer surface of the earbud shell. This resulted in having a continuous contact surface between the ear skin and the earbud. Secondly, the interconnection between the PCB and the PPG module was implemented with a FLEX board that provided some degrees of freedom in the mechanical integration. Finally, the size of the battery is found to be a trade-off between having an off-shelf part and a custom-shaped one, which directly affect the minimum size of the earbud. Referring to Fig. 13, the ergonomics of the earbud are primarily dependent upon the head diameter, d and the angle θ . These two parameters were set at 16.5 mm and 54° which allowed to fit all the population in our study, described in Section V. Secondly, the tail length is dependent upon the battery size and position of the PCB board within the earbud. According to previous experimental observations, the sensor location was found to be optimal at a equal to 6.75 mm, with respect to the AA plane. This was observed with respect to optimal PPG signal quality onto the tragus. The overall length of the earbud (h) is equal to 50 mm. Some functional prototypes, shown in Fig. 14, are 3D printed in stereolithography because of its intrinsic advantages, such as being cost-effective and having greater resolution for smaller features. A nylon-based material, EOS PA2200®, was selected because robust, flexible, stable for long periods of time and biocompatible under EN ISO 10993-1.

Fig. 14 shows two versions of the prototyped ear worn devices. One is targeting the PPG signal at the tragus of the ear and the second targets the PPG on the concha. Fig. 15 shows the PPG signal, measured on a same subject, using the tragus and concha prototypes. A fairly high SNR of 54 dB, on the raw data, is obtained at both ear locations while consuming



only 70 μA at the optical module level (LED and sensor), at a sampling rate of 120Hz. The prototype wearable earbud led to further ex-vivo studies of the PPG signal at the tragus and concha of the ear.

The ear-wearable system presented above relays on an optical module and monolithic sensor chip featuring low power and small form factor. In order to demonstrate that these advantages do not come at the cost of the PPG signal quality and its relevance for medical applications, a study performed on several subject is conducted to compare the precision of heart beat detection using the presented buds with a medical ECG.



A. Importance of Accurate Beat Detection

B. Beat Detection Using the Earbud

In what follows, we present a study that we performed to examine the beat-to-beat detection accuracy of using the PPG signal recorded by the presented earbud. Seven male subjects with a mean age of 34.29 ± 5.28 years participate in

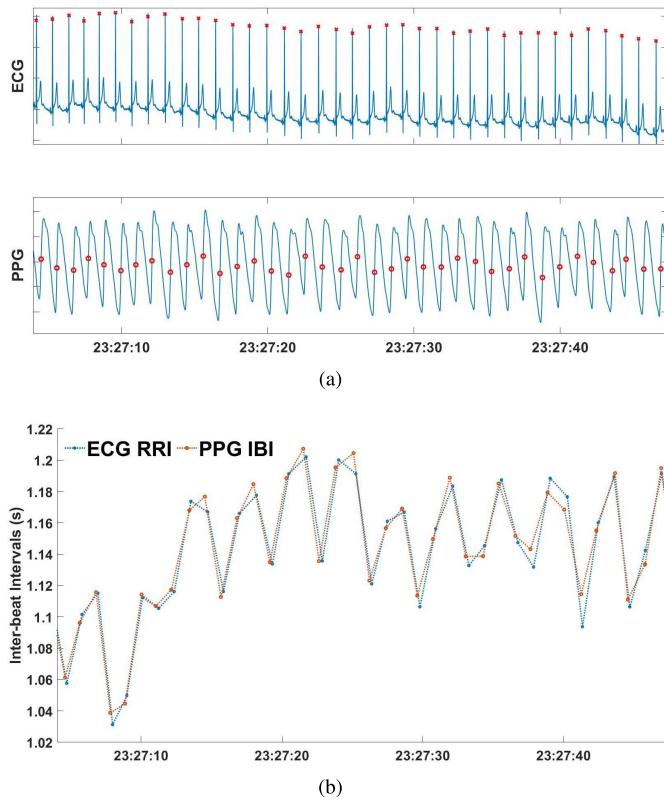


Fig. 16. (a) A plot of ECG and PPG signals with R-peak and maximum up-slope detection (red labels), respectively. (b) Corresponding ECG-based RR intervals and PPG-based IBI's.

the study. In total, 43 day and night recordings are performed with a total duration of 72.21 hours: 37.10 hours of *sleep* and 35.11 hours of *wake* recording. All *wake* recordings are done during working hours with as little motion as possible, while the *sleep* recordings are done during sleep. The participants are asked to wear the earbud in the left ear. Although we expected that the earbud gets displaced during the night recordings, it remained intact in all the recordings ensuring a good quality PPG signal. The sampling frequency of the earbud is set to 122 Hz and the driving current of the LED is set to 4.1 mA. For the reference ECG recordings, the Shimmer3 Consensus ECG development kit is used to monitor 4 ECG channels. The sampling frequency for the ECG recordings is set to 1024 Hz.

To extract the inter-beats interval (IBI) values from the PPG signals, we use the Senbiosys proprietary IBI detection software [35]. Note that the IBI detection and estimation algorithm is ran offline on the raw PPG data that is sent to the computer via bluetooth low energy (BLE). On the other hand, to extract the RR intervals (RRI) from the ECG signals, we use the ConsensusPRO Software version 1.6.0. In Fig. 16(a), we present plots of ECG and PPG recordings. The plots include (in red labels) the detection of R-peaks of the ECG signal and the detection of the maximum up-slopes of the PPG signal. These points are used to generate the ECG-based RR intervals and the PPG-based IBI values, depicted in Fig. 16(b). Further technical details regarding the algorithm can be found in [35].

In Table II, we summarize the statistics of the reference RRI values. The mean RRI value is 1.04 ± 0.17 s, which means that

TABLE II
THE STATISTICS OF THE REFERENCE RR INTERVALS

Number of RRI's	213072
Mean $\pm$ SD (ms)	1040 ± 170
pNN50 (%)	29.40
pNN20 (%)	71.13

TABLE III
THE BEAT DETECTION PERFORMANCE

Correctly Detected Beats	209811
Correct Beats (%)	98.47
Missed Beats (%)	1.53
Extra Beats (%)	1.84

on average the participants have HR values less than <60 beats per minute. Moreover, the average HRV of the participants can be considered to be fairly high with a pNN50 and pNN20¹ values of 29.40% and 71.13%, respectively.

C. Performance Evaluation

The performance of the IBI detection algorithm using the earbud is summarized in Table III. The classification of the beats as correct, missed, and extra beats is performed based on the technique proposed in [35]. We label each PPG beat as a correct beat, a missed beat, or an extra beat, based on the number of ECG beats detected in the vicinity of the PPG beat. We count the number of beats detected using the ECG signal in the interval $[t - 0.5l, t + 0.5l]$, where t is the time when the PPG beat was detected and l is the length of the corresponding IBI. If only one reference ECG beat is detected, then the corresponding PPG beat is labeled as a correct beat. If no reference beat is detected in the specified interval, then the corresponding PPG beat is labeled as an extra beat. All the ECG beats that are not referenced by a PPG beat are considered to be missed PPG beats. The results show that the percentage of the correctly detected beats is 98.47%. On the other hand, the percentage of the extra beats is only 1.84%.

For the IBI estimation, given that our recordings lack motion detection signals, such as triaxial acceleration signals, we use the following strategy to discard the noisy intervals. We divide the PPG recordings into 10-second intervals. Then, we keep the intervals where the algorithm correctly detects more than 80% of the reference beats. This is justified by the fact that intervals containing many missed or extra beats also contain, with high probability, motion artifacts. These artifacts may result not only from the mobility of the forearm, but also from very minor finger movements. For accurate IBI detection performance analysis and statistics computation, we therefore remove the intervals containing noisy signals. Upon discarding, 166275 beats are retained ($\sim 80\%$ of the all the detected beats) for the IBI estimation. In Table IV, we summarize the IBI estimation accuracy of the

¹pNN50 and pNN20 denote the percentage of the RR intervals with a successive difference exceeding 50 ms and 20 ms, respectively.

TABLE IV
THE PERFORMANCE OF THE IBI ESTIMATION

MAE (ms)	7.52
ME (ms)	0.65
RMSE (ms)	13.71
MAPE (%)	0.74

TABLE V
THE BEAT DETECTION-ESTIMATION PERFORMANCE: WAKE VS SLEEP

	Wake	Sleep
Correct Beats (%)	97.49	99.21
Missed Beats (%)	2.51	0.79
Extra Beats (%)	2.68	1.21
MAE (ms)	8.23	7.01
ME (ms)	0.91	0.47
RMSE (ms)	16.14	12.00
MAPE (%)	0.77	0.72

PPG with a mean absolute error (MAE) of 7.52 ms, a mean error (ME) of 0.65 ms, a root mean square error (RMSE) of 13.71 ms, and a mean absolute percentage error (MAPE) of 0.74%. As shown in Table V, these beat detection and IBI estimation performance metrics improve for sleep/night recordings because of the absence of motion artifacts. The beat estimation performance obtained suggests the feasibility of performing HRV analysis using the proposed earbud.

VI. CONCLUSION

Continuous and accurate heart beat monitoring is achieved with the presented earbud while consuming only 60 μ A at the chip level and less than 10 μ A average LED current. This performance is obtained thanks to a multidisciplinary optimization. The device embeds a monolithic optical sensor exploiting CIS process and featuring 2.25 mm² silicon area, which is considerably smaller than state of the art PPG sensors. It encompasses dedicated pixels featuring an almost flat QE response on the whole visible range, while still exhibiting good NIR response. The dedicated pixels array is combined with low power design and CMOS integration to result into an ultra-low power and miniaturized sensor chip.

The advantages obtained thanks to the monolithic sensor integration are enhanced by an optimal optical module design maximizing the PI of the PPG signal. The latter is optimized thanks to optical simulations emulating how the LED light propagates in the human skin.

The performance is further enhanced by an ergonomic design of the earbud allowing better diffusion of the light through the ear skin on two possible ear positions, namely the tragus and the concha.

The earbud system exhibits high SNR PPG signal (over 50 dB) while consuming less than 70 μ A on both sensor and emitter side. The two ear locations, tragus and the concha, show similar performance.

The system is validated for precise heart beat detection by comparing the earbud to a medical ECG device. The validation is performed on 7 subjects for a total of 72 hours of recording,

distributed between wake and sleep time. 98.47 % of the beats are correctly detected versus an ECG gold standard and the IBI estimation from these recordings exhibits a MAE of 7.52 ms.

This proposed wearable optical heart beat monitoring system shows a promising potential for wellness and medical health monitoring thanks to the ease of use, non-evasiveness and long battery life. Further clinical studies shall be performed to bring this wearable technology to the medical health monitoring market. Future research can also focus on the comparison between the tragus and concha PPG on a larger population.

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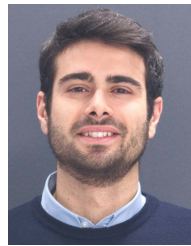


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